

CYBER INVESTIGATION SYSTEM

1. Abstract

With the rapid increase in cyber-crimes such as identity theft, online fraud, phishing, and data breaches, law enforcement agencies require efficient digital tools to manage and investigate suspect information.

This project presents a **Cyber Investigation System** that allows authorized personnel (e.g., inspectors) to securely log in and perform operations such as adding, searching, and managing suspect data.

The system incorporates features like **multi-field search, Aadhaar data masking, and search history tracking**, making it similar to real-world investigation tools such as CCTNS/NCRP.

2. Problem Statement

Traditional investigation systems face several challenges:

- a) Data is scattered across multiple sources
- b) Manual searching is time-consuming
- c) Weak authentication systems
- d) High risk of unauthorized access

This project aims to solve these problems by providing:

- ✓ Centralized database system
- ✓ Fast and efficient search functionality
- ✓ Secure authentication mechanism
- ✓ Automated investigation workflow

3. Objectives

- a) To develop a secure authentication system
- b) To build a structured suspect data management system
- c) To implement multi-parameter search functionality
- d) To protect sensitive data using masking techniques
- e) To maintain logs of investigation activities

4. Scope of the Project

- a) Can be used in cyber cells or police departments
- b) Useful for investigation and suspect tracking
- c) Suitable for academic and demonstration purposes
- d) Can be extended into large-scale systems like CCTNS

5. System Architecture

5.1 Architecture Type

This system follows a **3-Tier Architecture**:

1. Presentation Layer (Frontend)

- a) HTML-based user interface
- b) Handles user interaction

2. Application Layer (Backend)

- a) Developed using Flask (Python)
- b) Handles logic, routing, and processing

3. Data Layer (Database)

- a) MySQL database
- b) Stores and retrieves data

5.2 Data Flow

1. User submits login credentials
2. Flask verifies credentials from MySQL
3. On success, dashboard is displayed
4. User performs search or data entry
5. Query executed in database
6. Results displayed on UI (with masked data)

6. Technology Stack

Layer	Technology Used
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Frontend	HTML, CSS (basic)
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Backend	Python (Flask Framework)
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Database	MySQL
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Tools	VS Code, MySQL Workbench
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7. Functional Modules

7.1 Authentication Module

- a) Username and password-based login
- b) Session management
- c) Restricts unauthorized access

7.2 Dashboard Module

- a) Central interface for navigation
- b) Provides access to:
 - i. Search Suspect
 - ii. Add Record
 - iii. View History
 - iv. Logout

7.3 Data Entry Module

- a) Allows manual insertion of suspect records
- b) Fields include:
 - i. Name
 - ii. Father's Name
 - iii. Mobile Number
 - iv. Email ID
 - v. Aadhaar Number
 - vi. Address

7.4 Search Module (Core Feature)

Capabilities:

- a) Search using:
 - i. Mobile number
 - ii. Aadhaar number
 - iii. Name
 - iv. Father's name
 - v. Email
 - vi. Address

Search Type:

- a) Exact match
- b) Partial match using SQL LIKE operator

7.5 Data Masking Module

Sensitive information protection:

- a) Aadhaar number is partially hidden
- b) Only last 4 digits are visible

Example:

Original: 123456789012

Displayed: XXXX-XXXX-9012

7.6 Logging Module (Search History)

- a) Records every search performed
- b) Stores:
 - i. Username
 - ii. Search keyword
 - iii. Timestamp

7.7 Session Management

- a) Maintains login session
- b) Prevents unauthorized access to pages

8. Database Design**8.1 Users Table**

Field	Type
id	INT (PK)
username	VARCHAR
password	VARCHAR

8.2 Records Table

Field	Type
id	INT
name	VARCHAR

Field	Type
father_name	VARCHAR
mobile	VARCHAR
email	VARCHAR
Aadhar	VARCHAR
address	TEXT

8.3 Search Logs Table

Field	Type
id	INT
username	VARCHAR
keyword	VARCHAR
search_time	TIMESTAMP

9. System Workflow

1. User logs in using valid credentials
2. Dashboard is displayed
3. User selects an action:
 - a) Add new record
 - b) Search existing records
4. Search results are displayed
5. Search activity is logged
6. User logs out

10. Security Features

- a) Session-based authentication
- b) Route protection
- c) Input validation
- d) Aadhaar data masking
- e) Controlled database queries

11. Advantages

- ✓ Fast and efficient search
- ✓ Secure access control
- ✓ Easy-to-use interface
- ✓ Real-world simulation of investigation tools
- ✓ Scalable architecture

12. Limitations

- a) Basic UI design
- b) No encryption implemented
- c) No external API integration
- d) Not deployed online

13. Future Enhancements

- a) Role-based access control (Admin/Inspector)
- b) OTP-based authentication
- c) Data encryption
- d) Cloud deployment
- e) Integration with real-time APIs
- f) Advanced analytics dashboard

14. Testing

Test Cases:

- a) Valid login → Successful access
- b) Invalid login → Error message
- c) Add record → Data stored successfully
- d) Search valid data → Results displayed
- e) Search invalid data → No result

15. Conclusion

The Cyber Investigation System is a functional prototype that simulates real-world investigation workflows.

It successfully integrates:

- a) Secure authentication
- b) Data management
- c) Smart search functionality
- d) Data masking
- e) Activity tracking

This makes it a strong foundation for developing advanced cyber investigation platforms.

16. Developer Details

Name: Vicky

Role: Developer

Technology Stack: Python, Flask, MySQL